Interpolation with Fuzzy Input and Output and Spline Function: Application to Diffusive Models

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***Abstract*—This study investigates the interpolation of uncertain data represented by linearly correlated fuzzy numbers. To this end, we utilize the algebraic structure of the space $\mathbb{R}_{\mathcal{F}(A)}$, which is isometric and isomorphic to the complex field. We demonstrate the potential of our approach to fuzzy polynomial and spline interpolation methods by applying it to diffusive models involving impedance.**

***Index Terms*—Fuzzy numbers, fuzzy interpolation, polynomial approximation, spline interpolation, uncertain data modeling, fuzzy algebraic structures, admittance modeling.**

I. Introduction

Fuzzy interpolation has been studied by many researchers, with most approaches focusing on obtaining fuzzy number-valued functions defined over an interval of the real numbers (see e.g., [1]–[3] and the references therein). Approaches based on Zadeh's extension principle for interpolating pairs of fuzzy numbers are presented in [4], [5]. This paper presents a fuzzy polynomial method to interpolate pairs of uncertain data, where both the inputs and outputs are modeled by linearly correlbated fuzzy numbers ($\mathbb{R}_{\mathcal{F}(A)}$). Additionally, it explores a spline method in which only the outputs are modeled by linearly correlated fuzzy numbers. Under certain mild conditions on the fuzzy number A, this space is a Banach space [6]. In contrast to the aforementioned approaches based on Zadeh's extension principle, our method leverages the structure of the Banach space $\mathbb{R}_{\mathcal{F}(A)}$. To demonstrate the potential of our approach, we apply the proposed polynomial and spline interpolation methods to estimate admittance in diffusive models [7]. In general, the mathematical modeling of admittance phenomena in diffusive systems heavily involves the algebraic structure of complex numbers. This is one of the motivations behind our choice, as $\mathbb{R}_{\mathcal{F}(A)}$ is isometric and isomorphic to the field of complex numbers $\mathbb{C}$ [8].

II. Preliminary

Let $\mathbb{R}_{\mathcal{F}}$ be the class of fuzzy numbers. The membership function of $A \in \mathbb{R}_{\mathcal{F}}$ is denoted by μ_A and its α-levels are given by $[A]_\alpha = [a_\alpha^-, a_\alpha^+], \forall \alpha \in [0,1]$. A triangular fuzzy number A is denoted by the triple $(a; b; c)$, with $a \leqslant b \leqslant c$, where its α-levels are given by $[a + \alpha(b-a), c - \alpha(c-b)], \forall \alpha \in [0,1]$.

A fuzzy number A is said to be symmetric with respect to $x \in \mathbb{R}$ if $\mu_A(x-y) = \mu_A(x+y)$, $\forall y \in \mathbb{R}$. A fuzzy number A is said to be asymmetric if there exists no x such that A is symmetric.

A. The space of A-Linearly Correlated Fuzzy Numbers

Consider A in $\mathbb{R}_{\mathcal{F}}$ and $\Phi : \mathbb{C} \to \mathbb{R}_{\mathcal{F}}$ be the operator which maps each $(r + iq) \in \mathbb{C}$ to the fuzzy number $\Phi(r + iq)$ given by $\Phi(r + iq) = r + qA,\ \forall r, q \in \mathbb{R}$. The range of this operator is denoted by $\mathbb{R}_{\mathcal{F}(A)}$.

Theorem 1. *[6], [8] Let $A \in \mathbb{R}_{\mathcal{F}}$. The operator $\Phi : \mathbb{C} \to \mathbb{R}_{\mathcal{F}(A)}$ given by*

$$\Phi(r + iq) = r + qA, \tag{1}$$

for all $z = (r + iq) \in \mathbb{C}$ is a bijection if, and only if, A is asymmetric.

Roughly speaking, the space $\mathbb{R}_{\mathcal{F}(A)}$ is obtained from $\mathbb{C}$ simply replacing the imaginary unit i by the asymmetric fuzzy number A. However, unlike the complex case, the quantity $r + qA$ can be seen as a sum of fuzzy numbers r and qA that results in a fuzzy number $B = r + qA$. Since r represents a real number and qA is a non-crip fuzzy number, we define $r = \mathrm{Re}(B)$ and $q = \mathrm{Fu}(B)$.

An important fact that is worth emphasizing is that $\mathbb{R}$ can be embedded $\mathbb{R}_{\mathcal{F}(A)}$, since every real number r can be identified with the fuzzy number $\Phi(r + i0) \in \mathbb{R}_{\mathcal{F}(A)}$. Another interesting property is that for every $B \in \mathbb{R}_{\mathcal{F}(A)} \backslash \mathbb{R}$, $\mathbb{R}_{\mathcal{F}(B)} = \mathbb{R}_{\mathcal{F}(A)}$ [6]. In other words, the range of Φ does not depend on the choice of $A \in \mathbb{R}_{\mathcal{F}(A)} \backslash \mathbb{R}$.

Theorem 2. *[8] Let A be an asymmetric fuzzy number. The set $\mathbb{R}_{\mathcal{F}(A)}$ is field isomorphic to $\mathbb{C}$ where the arithmetic operations are given by*

$$B \circledast C = \Phi(\Phi^{-1}(B) * \Phi^{-1}(C)),$$

for every $B, C \in \mathbb{R}_{\mathcal{F}(A)}$, where $*$ stands one of the four arithmetic operations of $\mathbb{C}$, i.e., $* \in \{+, -, \cdot, \div\}$.

In addition, $\mathbb{R}_{\mathcal{F}(A)}$ is isometric to $\mathbb{C}$ with the norm

$$\|B\|_\Phi = |\Phi^{-1}(B)| = \sqrt{r^2 + q^2}. \tag{2}$$

From now on, unless otherwise stated, we assume that the fuzzy number A that generates the space $\mathbb{R}_{\mathcal{F}(A)}$ is asymmetric and, therefore, $\mathbb{R}_{\mathcal{F}(A)}$ is a field according to Theorem 2.

Through the isomorphism (1), the arithmetic operators of the fuzzy numbers $B = r + qA$ and $C = s + pA$ can be rewritten as follows

$$B \oplus C = (r + s) + (q + p)A, \tag{3}$$

$$B \ominus C = (r - s) + (q - p)A, \tag{4}$$

$$B \odot C = (rs - qp) + (sq + rp)A, \tag{5}$$

$$B \oslash C = \frac{rs + qp}{s^2 + p^2} + \frac{qs - rp}{s^2 + p^2}A, \text{ for } C \neq 0. \tag{6}$$

One can also use the notation $\frac{B}{C}$ instead of $B \oslash C$.

III. Polynomial and Spline Interpolation in $\mathbb{R}_{\mathcal{F}(A)}$

Next, we will briefly present a study for polynomial and spline interpolation. This study is a consequence of the good properties of the $\mathbb{R}_{\mathcal{F}(A)}$ field.

A. Polynomial Interpolation in $\mathbb{R}_{\mathcal{F}(A)}$

Polynomial interpolation via Lagrange polynomial is a technique used to find a polynomial that passes through a given set of points

Theorem 3. *Let the data set $D = \{(Z_i, Y_i) \in \mathbb{R}^2_{\mathcal{F}(A)} \mid i = 0, \ldots, n\}$ be such that $Z_i \neq Z_j$ for $i \neq j$. The fuzzy polynomial $p_n : \mathbb{R}_{\mathcal{F}(A)} \to \mathbb{R}_{\mathcal{F}(A)}$, of degree less or equal n, given by (the Lagrange form)*

$$p_n(Z) = \sum_{k=0}^{n} Y_k \odot L_k(Z), \tag{7}$$

where L_k is kth-Lagrange polynomial given by

$$L_k(Z) = \frac{\Pi_{j \neq k}(Z \ominus Z_j)}{\Pi_{j \neq k}(Z_k \ominus Z_j)} \tag{8}$$

for $k = 0, \ldots, n$, is such that $p_n(Z_i) = Y_i$ for all $i = 0, \ldots, n$. That is, it interpolates the data set.

Proof. Consider the basis polynomial $L_k(Z_i)$. Thus, $L_k(Z_i) = 0$, for $i \neq k$, and $L_k(Z_i) = 1$, for $i = k$. Therefore,

$$p_n(Z_i) = Y_i \odot L_i(Z_i) \oplus \sum_{\substack{k=0 \\ k \neq i}}^{n} Y_k \odot L_k(Z_i) = Y_i \cdot 1 \oplus \sum_{\substack{k=0 \\ k \neq i}}^{n} Y_j \cdot 0 = Y_i.$$

□

B. Spline in $\mathbb{R}_{\mathcal{F}(A)}$

Consider $f : \mathbb{R} \to \mathbb{R}_{\mathcal{F}(A)}$ a function be known in the points (x_i, Y_i), with $Y_i = f(x_i)$ for $i = 0, \ldots, n$. An alternative to fuzzy polynomial interpolation is to use spline functions, where the idea is to estimate the function f in each interval $[x_{i-1}, x_i]$ by polynomials that satisfy certain boundary conditions as follows.

Definition 1. *Consider a function $f : \mathbb{R} \to \mathbb{R}_{\mathcal{F}(A)}$ and $x_0 < x_1 < \cdots < x_n$, with $x_i \in \mathbb{R}, i = 0, 1, \cdots, n$. A function $S_p : [x_0, x_n] \to \mathbb{R}_{\mathcal{F}(A)}$ is called fuzzy interpolating spline of degree p with nodes at points $x_i, i = 0, 1, \cdots, n$ if the following conditions are satisfied*

a) *At each interval $[x_{i-1}, x_i], i = 1, \cdots, n$, $S_p(x)$ is a polynomial of degree p: $s_i(x)$;*
b) *$S_p(x)$ is continuous and has a continuous derivative up to order $(p - 1)$;*
c) *$S_p(x_i) = f(x_i)$, $i = 0, 1, \cdots, n$.*

Since $f : \mathbb{R} \to \mathbb{R}_{\mathcal{F}(A)}$ is given by $f(x) = r(x) + q(x)A$, we obtain that

$$S_p(x) = \text{Spline}(r(x)) + \text{Spline}(q(x))A. \tag{9}$$

IV. Diffusive Models

Impedance or admittance spectroscopy (EIS or EAS) consists of determining the impedance or admittance of a given system. For Barsoukov and Macdonald [9], the impedance results from the combined effects of resistance and reactance (capacitive and inductive) when the system is subjected to a time-varying alternating current or potential.

According to [7], impedance can be characterized using a Diffusive Model, which treat the electrode and electrolyte as continuous media, where mobile species transport is governed by differential equations. The charge transfer across interfaces follows reaction rate laws, which act as boundary conditions. Usually, impedance is expressed as a complex-valued function $Z : \mathbb{R} \to \mathbb{C}$

$$Z(\omega) = R(\omega) + jX(\omega), \tag{10}$$

where $\omega = 2\pi f$ (rad/s) is the angular frequency, f is the frequency in Hertz (Hz), j the imaginary unit, R the real part (related to energy dissipation, that is, resistance), and X the imaginary part (reactance, representing energy storage in capacitors or inductors). In SI units, impedance is measured in Ohms (Ω).

Immittances quantities, such as admittance, can be derived from impedance [9]. The admittance function $Y : \mathbb{C} \to \mathbb{C}$ (whose the unit is siemens (S)), quantifies how easily a circuit/component allows current flow when a voltage is applied. It is the reciprocal of impedance

$$Y(R+jX) = \frac{1}{R + jX} = \frac{R}{R^2 + X^2} - j\frac{X}{R^2 + X^2} = G + jB, \tag{11}$$

where the real part G is called conductance and measures the ability to conduct current, and the imaginary part B is called susceptance and represents the reactive (capacitive/inductive) behavior.

In $\mathbb{R}_{\mathcal{F}(A)}$, using isometric isomorphism (1), the fuzzy impedance $\tilde{Z} = \Phi \circ Z : \mathbb{R} \to \mathbb{R}_{\mathcal{F}(A)}$ is given by

$$\tilde{Z}(\omega) = R(\omega) + X(\omega)A, \tag{12}$$

and, in this case, X is called fuzzy part. Furthermore, fuzzy admittance $\tilde{Y} = \Phi \circ Y \circ \Phi^{-1} : \mathbb{R}_{\mathcal{F}(A)} \to \mathbb{R}_{\mathcal{F}(A)}$ is defined as

$$\tilde{Y}(Z) = \frac{1}{Z} = \frac{R}{R^2 + X^2} - \frac{X}{R^2 + X^2}A = G + BA. \tag{13}$$

One can also use the notation Z and Y instead of $\tilde{Z}$ and $\tilde{Y}$.

V. Application of Polynomial and Spline Interpolation in the Admittance

In this section, we estimate the admittance in a diffusive model using fuzzy Polynomial and Spline Interpolation. To this end, we consider a dataset composed by data of an electrochemical cell experiment for an anomalous model, which consists of an extension of the Poisson-Nernst-Planck model [7].

A. Polynomial Case

We use formula of the Lagrange polynomial to find a fuzzy polynomial of degree 4 that passes through five pairs $(Z_i, Y_i) \in \mathbb{R}^2_{\mathcal{F}(A)}$, excerpted from [7] according to Table I.

TABLE I
Real and fuzzy parts of Z and Y

R	X	G	B
18339.0	-301650	2.008014×10^{-7}	3.302892×10^{-6}
-11976.0	-269260	-1.648579×10^{-7}	3.706550×10^{-6}
-2844.7	-128640	-1.718193×10^{-7}	7.769832×10^{-6}
-2823.0	-53921	-9.682917×10^{-7}	1.849495×10^{-5}
1465.9	-17724	4.634684×10^{-6}	5.603735×10^{-5}

Using the general formula (7), we obtain that the polynomial $p_4 : \mathbb{R}_{\mathcal{F}(A)} \to \mathbb{R}_{\mathcal{F}(A)}$ of order 4 that is used to approximate the function (13) is given by

$$\begin{aligned} p_4(Z) = {} & (2.421726\times10^{-27} + 9.933194\times10^{-26}A) \odot Z^4 \\ & \oplus (-7.660953\times10^{20} + 1.652947\times10^{-21}A) \odot Z^3 \\ & \ominus (5.135835\times10^{-16} + 2.046060\times10^{-14}A) \odot Z^2 \\ & \oplus (2.209671\times10^{-9} - 7.479196\times10^{-11}A) \odot Z \\ & \oplus (3.530517\times10^{-6} + 8.931158\times10^{-5}A). \end{aligned} \tag{14}$$

One can express $p_4(Z)$ as a function of R (resistance) and X (reactance). That is, for $Z = R + XA$,

$$p_4(R + XA) = \mathrm{Re}(p_4(R + XA)) + \mathrm{Fu}(p_4(R + XA))A. \tag{15}$$

In $\mathbb{C}$, the real (conductance) and imaginary (susceptance) components of the admittance function are usually represented separately and in terms of frequency given in Hertz [10]. On the other hand, in $\mathbb{R}_{\mathcal{F}(A)}$, the susceptance is described by an uncertain curve, which is related to the reactive (capacitive/inductive) behavior of the circuit/component. Fig.1 shows the approximation for the fuzzy admittance which is the result of the addition of the effects of the conductance

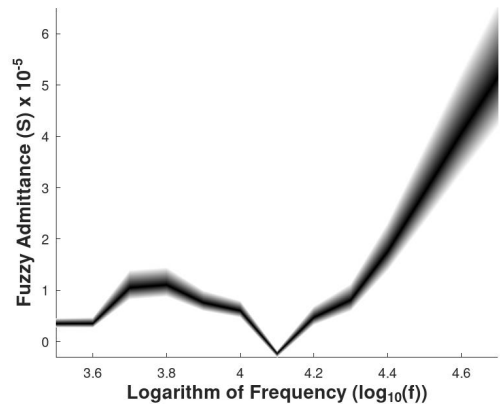

Fig. 1. Representation of the fuzzy Lagrange polynomial of degree 4 (or the fuzzy admittance), which represents the total effect of conductance with susceptance, with $A = (0.8; 1; 1.3)$. The darker the lines, the greater the membership degree of $R + XA$ in the fuzzy number $p_4(R + XA)$

(real part) and the susceptance (fuzzy part) of Eq. (13). Note that the conductance and susceptance have the same unit (S) and effect on the phenomenon. Thus, the fuzzy approach enables a joint visualization of conductance and susceptance, representing the real and fuzzy components, respectively, with respect to frequency, offering a broader perspective on system behavior. Rather than analyzing the real (conductance) and fuzzy (susceptance) curves in isolation, combining these components allows us to explore their interaction, which is essential for modeling physical phenomena. This integrated view provides a unified representation of both crisp values and their associated uncertainties, which is particularly valuable in contexts involving imprecise data or dynamic variability. Here, the choice of the triangular fuzzy number $A = (0.8; 1; 1.3)$ is purely theoretical and justified by the need to take into account uncertainties and tolerances inherent in the experimental data. Specifically, the values $a = 0.8$ and $c = 1.3$ represent realistic deviations around the value $b = 1$, encompassing potential measurement errors or dynamic variations in the system.

With the fuzzy approach, one can identify regions in the domain where uncertainty around admittance is greater (or smaller). This may drive frequency regions of interest. For example, one can observe in Fig.1 that the uncertainty in the admittance is increasing for frequencies f such that $log_{10}(f) > 4.3$, that is, for frequencies above 1.9×10^4 Hz.

In the following subsection, we continue our study on admittance using fuzzy interpolating spline for the data set contained in [7]. For this purpose, we selected additional data with the highest variability in admittance.

B. Cubic Spline Case

This subsection presents an approximation to the admittance Y in terms of spline interpolation. To this end, from Definition 1, the admittance must be regarded as a function whose its domain is an interval of the real line. Thus, we consider the function $Y : \mathbb{R} \to \mathbb{R}_{\mathcal{F}(A)}$ given by $Y(\omega) = 1/Z(\omega)$ and the dataset in the form $(\omega_i, Y_i) \in \mathbb{R} \times \mathbb{R}_{\mathcal{F}(A)}$.

The fuzzy cubic spline S_3 has continuous first and second derivatives, which ensures that S_3 is a smooth curve without sharp peaks or abrupt changes in curvature at the nodes. Each segment, $s_i(\omega)$, is a cubic polynomial defined as follows

$$s_i(\omega) = a_i(\omega - \omega_i)^3 \oplus b_i(\omega - \omega_i)^2 \oplus c_i(\omega - \omega_i) \oplus d_i, \quad (16)$$

where $\omega \in [\omega_{i-1}, \omega_i]$ for $i = 1, \ldots, n$, and $a_i, b_i, c_i, d_i \in \mathbb{R}_{\mathcal{F}(A)}$ and can be represented in terms of its real and fuzzy parts

$$\text{Re}(s_i(\omega)) = \alpha_i(\omega - \omega_i)^3 + \gamma_i(\omega - \omega_i)^2 + \kappa_i(\omega - \omega_i) + \eta_i,$$

and

$$\text{Fu}(s_i(\omega)) = \beta_i(\omega - \omega_i)^3 + \delta_i(\omega - \omega_i)^2 + \zeta_i(\omega - \omega_i) + \theta_i,$$

where $a_i = \alpha_i + \beta_i A$, $b_i = \gamma_i + \delta_i A$, $c_i = \kappa_i + \zeta_i A$, $d_i = \eta_i + \theta_i A$ and $s_i(\omega) = \text{Spline}(\text{Re}(s_i(\omega))) + \text{Spline}(\text{Fu}(s_i(\omega)))A$.

The computation of $S_3(\omega)$ requires determining $4n$ fuzzy coefficients. Specifically, the fuzzy cubic spline $S_3(\omega)$ is defined by the following conditions

a) $S_3(\omega) = s_i(\omega)$ if $\omega \in [\omega_{i-1}, \omega_i]$;
b) $S_3(\omega_i) = Y(\omega_i), \quad i = 0, \ldots, n$;
c) $s_i(\omega_i) = s_{i+1}(\omega_i), \quad i = 1, \ldots, (n-1)$;
d) $s_i'(\omega_i) = s_{i+1}'(\omega_i), \quad i = 1, \ldots, (n-1)$;
e) $s_i''(\omega_i) = s_{i+1}''(\omega_i), \quad i = 1, \ldots, (n-1)$.

For natural splines, the second derivative at the endpoints is set to zero, specifically $s_1''(\omega_0) = 0$ and $s_n''(\omega_n) = 0$. These two additional conditions complete the system, yielding the $4n$ equations required to determine the fuzzy coefficients.

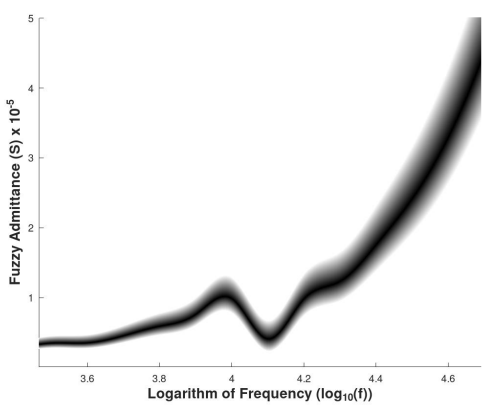

Fig. 2. Fuzzy interpolating cubic spline with $A = (0.8; 1; 1.3)$. The darker the lines, the greater the membership degree of ω in the fuzzy number $S_3(\omega)$

Fig.2 depicts the approximation of $Y(\omega)$ given by the cubic spline. It can be seen that the uncertainty or variability (evaluated by the diameter of the support $[S_3(Z)]_0$) of the admittance becomes more noticeable for values of $\log_{10}(f) > 4.1$, corresponding to frequencies above 1.2×10^4 Hz.

VI. Conclusions

In this study, we investigate polynomial interpolation of data with both uncertain inputs and outputs, as well as spline interpolation with uncertain outputs. To this end, we consider the class of A-linear fuzzy numbers ($\mathbb{R}_{\mathcal{F}(A)}$), which is isomorphic to the complex number field $\mathbb{C}$ when A satisfies the mild condition of being an asymmetric fuzzy number. To illustrate our approach, we study impedance in diffusive models focusing on admittance, which is a quantity that derives from impedance. This phenomenon is traditionally studied in the complex domain $\mathbb{C}$, where the real and imaginary components, called conductance and susceptance, respectively, are analyzed separately [10]. However, it is worth noting that conductance and susceptance have the same unit, nature, and impact on the phenomenon. Therefore, to combine both quantities into a single numerical value, one may consider the modulus of admittance [7], [11]. In contrast, in $\mathbb{R}_{\mathcal{F}(A)}$, each element represents a unique fuzzy number, which can be interpreted as the sum of a real part and a fuzzy part. For example, in Fig.2, for the value $\log_{10}(f) = 4$ (i.e., the frequency $f = 10^4$ Hz), the admittance $-2.91 \times 10^{-7} + 9.95 \times 10^{-6}(0.8; 1; 1.3)$ is equal to the fuzzy number $(7.67 \times 10^{-6}; 9.66 \times 10^{-6}; 1.26 \times 10^{-5})$. From the modeling point of view, this can seem as a gain since one can interpret this result as noise around the sum of conductance and susceptance. This type of analysis is only possible in $\mathbb{R}_{\mathcal{F}(A)}$, where conductance and susceptance can be represented and interpreted together as a single fuzzy number.

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